

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- **Data collection:** SpaceX REST API + web scraping of Wikipedia (using Python BeautifulSoup library)
- **Data wrangling:** filtered data to Falcon 9, filled missing payload mass, created the landing label
- **Exploratory data analysis:** seaborn/matplotlib charts and SQL queries
- **Visuals:** Folium maps and Plotly Dash dashboard
- **Predictive analysis:** Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV

Summary of results

- 66% overall first-stage landing success
- 83% test accuracy of the best model
- Success rate climbed at a steady rate from 2013 to 2020
- ES-L1, GEO, HEO and SSO orbits: 100% success
- KSC LC-39A has the most successful launches
- Factors that matter most: flight number (experience), orbit, launch site and payload mass

Introduction

Project background and context

SpaceX advertises Falcon 9 launches at about $62 million, while other providers charge upward of $165 million. Most of the savings come from reusing the first stage.

If we can predict whether the first stage will land successfully, we can estimate the cost of a launch.

Problems we want to answer

- Which factors (payload mass, orbit, launch site, flight number, booster version, etc.) influence most whether the first stage lands?
- How do those factors interact with each other?
- Where are the launch sites, and what is around them?
- Can a machine learning model predict a successful landing, and which model does it best?

Section 1

Methodology



Methodology

Executive summary

- **Data collection methodology:** requested launch data from the SpaceX REST API and scraped Falcon 9 launch tables from Wikipedia
- **Data wrangling:** kept Falcon 9 launches only, replaced missing payload mass with the mean, and converted landing outcomes into a 0/1 label for training
- **Exploratory data analysis (EDA):** scatter, bar and line charts with seaborn/matplotlib + SQL queries
- **Interactive visual analytics:** Folium maps of launch sites and outcomes; Plotly Dash dashboard with pie chart and payload scatter
- **Predictive analysis:** standardized features, split train/test, tuned four classifiers with GridSearchCV, compared accuracy and confusion matrices

GitHub repository: https://github.com/dorotheemorandgrondin/applied-datascience-training

Data Collection

Data was collected from two sources and combined into one dataset:

- **SpaceX REST API:** GET requests to api.spacexdata.com/v4/launches/past, then extra calls by ID for rockets, launchpads, payloads and cores
- **Wikipedia:** web scraping of the 'List of Falcon 9 and Falcon Heavy launches' page (9 June 2021 snapshot) with BeautifulSoup

Data Collection – SpaceX API

- Sent a GET request to the SpaceX launches endpoint and checked status code 200
- Decoded the JSON with .json() and flattened it with pd.json_normalize()
- Kept rocket, payloads, launchpad, cores, flight number and date; dropped multi-core and multi-payload launches; limited dates to 13 Nov 2020
- Used helper functions to call the API by ID: getBoosterVersion, getLaunchSite, getPayloadData, getCoreData
- Built a DataFrame from the dictionary, kept Falcon 9 only (90 launches), and replaced missing PayloadMass with the mean

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

- Requested the Wikipedia page (9 June 2021 snapshot) with requests.get()
- Created a BeautifulSoup object from the HTML response
- Found all tables; the launch records start at the third table
- Extracted column names from the <th> header cells
- Looped over each table row foro flight number, date, time, booster version, launch site, payload, mass, orbit, customer, launch outcome and booster landing
- Converted the dictionary to a DataFrame and saved it as CSV

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

- Calculated the percentage of missing values and identified numerical vs. categorical columns
- Counted launches per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit (GTO 27, ISS 21, VLEO 14, …)
- Counted landing outcomes (e.g. True ASDS 41, None None 19, True RTLS 14) and grouped the unsuccessful ones as bad_outcomes
- Created the Class label: 0 = the outcome is in bad_outcomes, 1 otherwise. Average success rate = 0.667 (66%)

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/labs-jupyter-spacex-Data_wrangling.ipynb

EDA with Data Visualization

Chart	Type	Rationale
Flight Number vs. Payload Mass	Scatter (catplot)	See how experience and payload together affect landing
Flight Number vs. Launch Site	Scatter (catplot)	Compare success over time at each site
Payload Mass vs. Launch Site	Scatter (catplot)	Check which sites handle heavy payloads and how they fare
Success Rate vs. Orbit Type	Bar chart	Compare the average success rate of each orbit
Flight Number vs. Orbit Type	Scatter (catplot)	See whether success in an orbit improves with experience
Payload Mass vs. Orbit Type	Scatter (catplot)	See how payload affects landing for each orbit
Yearly Launch Success Trend	Line chart	Show how the success rate changed over the years

Then performed feature engineering: selected 12 features, one-hot encoded Orbit, LaunchSite, LandingPad and Serial with get_dummies(), and cast everything to float64 (dataset_part_3.csv).

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/jupyter-labs-eda-dataviz.ipynb

EDA with SQL

- Displayed the names of the unique launch sites
- booster version and launch site
- Ranked the count of landing outcomes between 201Displayed 5 records where the launch site begins with 'CCA'
- Calculated the total payload mass carried by boosters launched by NASA (CRS)
- Calculated the average payload mass carried by booster version F9 v1.1
- Found the date of the first successful landing on a ground pad
- Listed boosters with successful drone ship landings and payload between 4,000 and 6,000 kg
- Counted the total number of successful and failed mission outcomes
- Listed the booster versions that carried the maximum payload mass (subquery)
- Listed 2015 drone ship landing failures with month, 0-06-04 and 2017-03-20

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map object	Content	Rationale
folium.Circle + Marker (label)	Each launch site's location and name	See where the sites are relative to the equator and coast
MarkerCluster with green/red markers	Every launch, coloured by outcome (green = success, red = failure)	Compare success rates between sites at a glance
MousePosition	Lat/long under the mouse pointer	Find coordinates of coastlines, railways, highways and cities
Marker with distance (DivIcon)	Distance in km from site to a nearby point	Measure how close sites are to infrastructure
folium.PolyLine	Line from a launch site to the point of interest	Visualize those proximity distances on the map

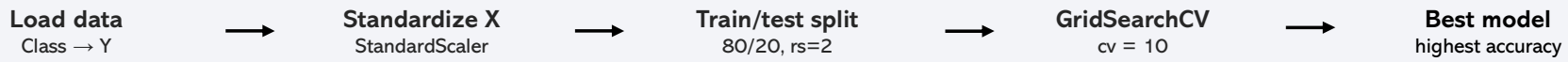
GitHub notebook: https://github.com/dorotheemorandgrondin/DataScienceCourse /blob/main/lab-jupyter-launch-site-location.ipynb

Build a Dashboard with Plotly Dash

Element	Content	Rationale
Launch site dropdown	Choose All Sites or one specific site	Lets us focus on one site or compare all of them
Pie chart (success)	All sites: share of successful launches per site. One site: success vs. failure	Shows which site contributes most successes and each site's success ratio
Payload range slider	Filter launches by payload mass (0–10,000 kg)	Explore how payload mass changes the outcome
Scatter plot (payload vs. outcome)	Payload mass vs. class, coloured by booster version	Spot which payload ranges and booster versions succeed most

GitHub Dash app: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)

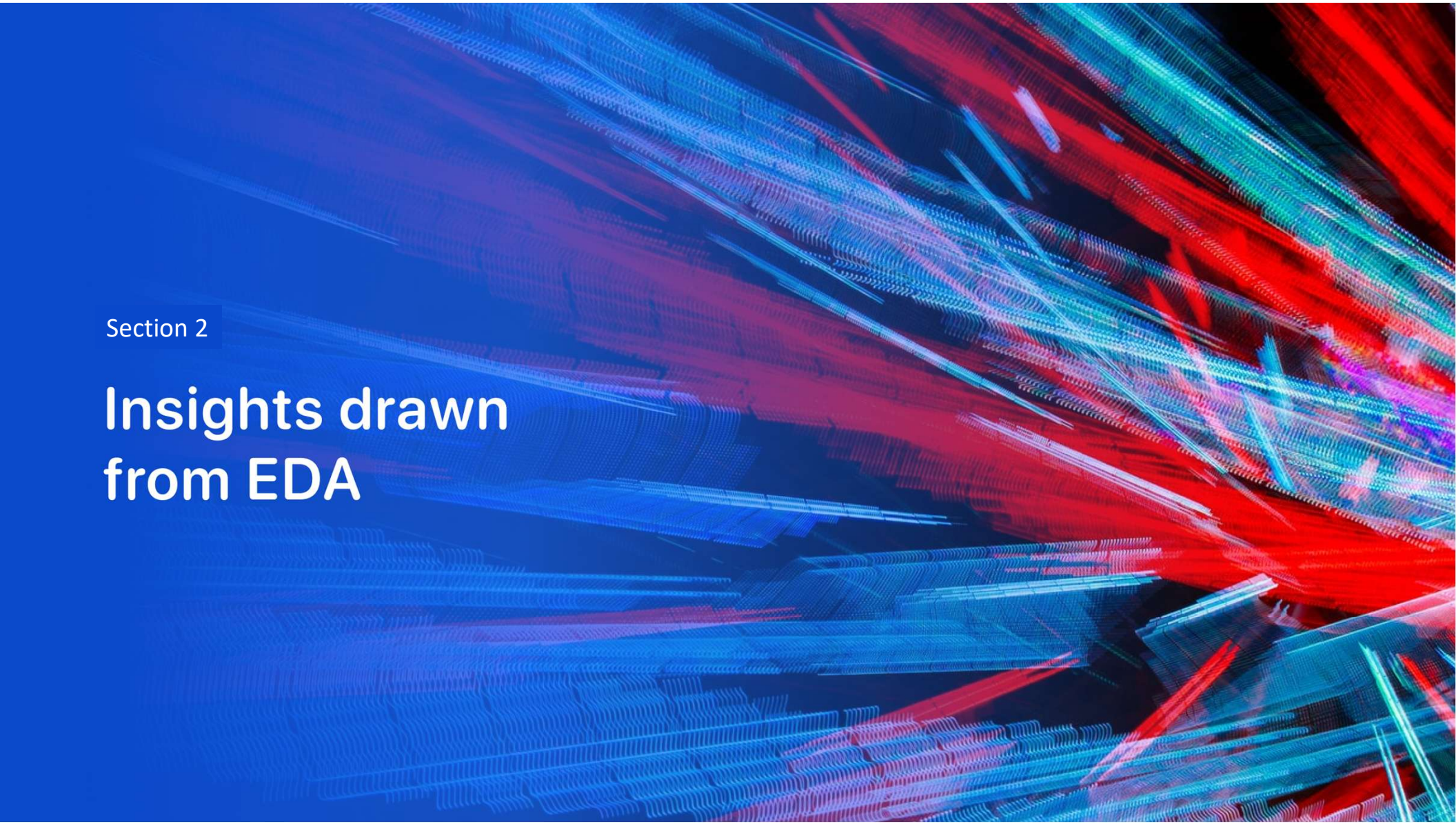


- **Build:** created Y from the Class column and standardized the 83 one-hot encoded features
- **Split:** train_test_split with test_size = 0.2 → 72 training and 18 test samples
- **Tune:** GridSearchCV (10-fold) on Logistic Regression, SVM, Decision Tree and KNN to find the best hyperparameters
- **Evaluate:** compared validation accuracy (best_score_), test accuracy (.score) and confusion matrices
- **Select:** picked the model with the highest accuracy

GitHub notebook: https://github.com/dorotheemorandgrondin/applied-datascience-training/blob/main/SpaceX-Machine-Learning-Prediction-Part-5.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

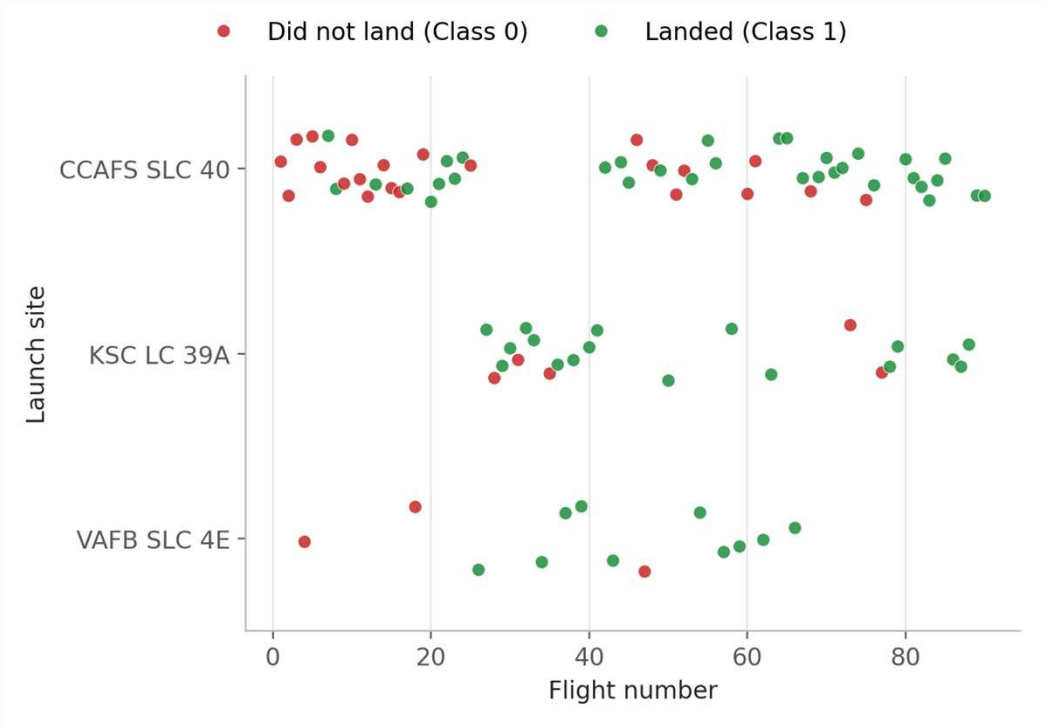


Section 2

Insights drawn from EDA

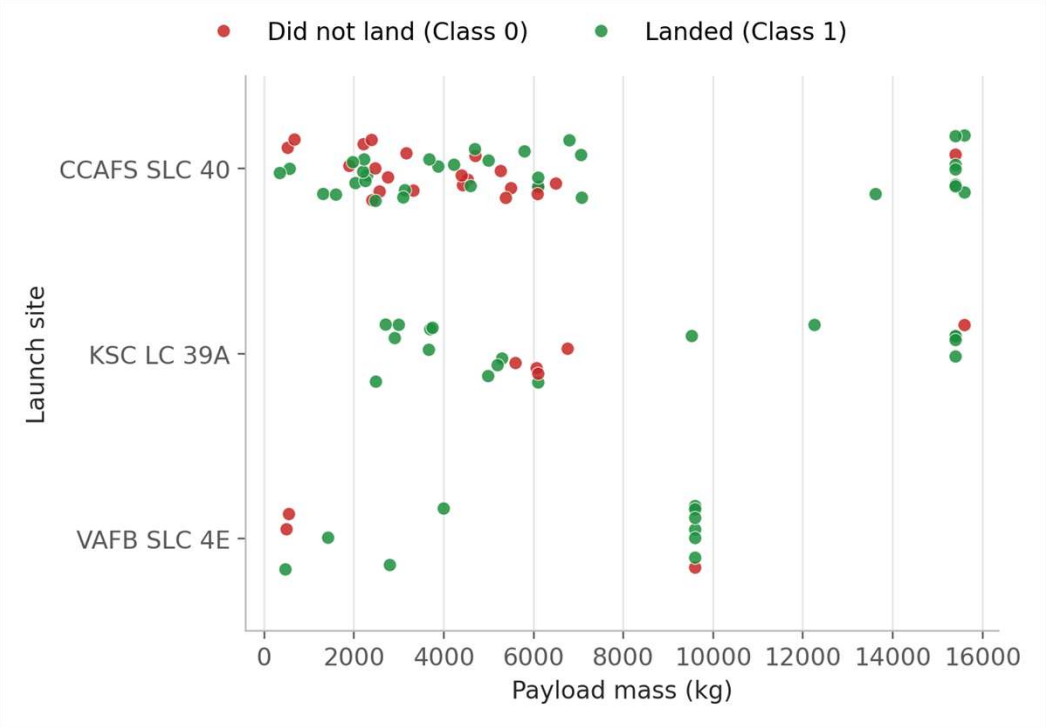
Flight Number vs. Launch Site

- Success becomes more frequent as flight number grows: early flights mostly failed, later ones mostly landed
- CCAFS SLC 40 has the most launches, including most of the early failures
- KSC LC 39A and VAFB SLC 4E show high success rates for later flights



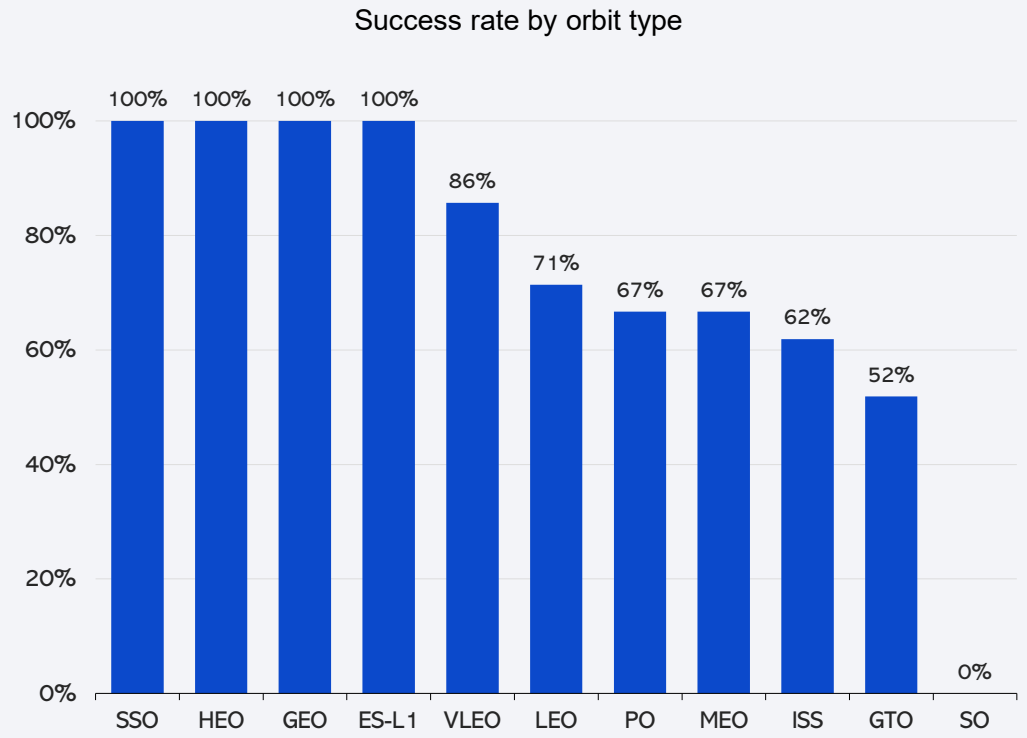
Payload vs. Launch Site

- VAFB SLC 4E launched no heavy payloads (greater than 10,000 kg)
- Heavy payloads above 8,000 kg have a high success rate, mostly from CCAFS SLC 40 and KSC LC 39A
- Lighter payloads show a mix of successes and failures at every site



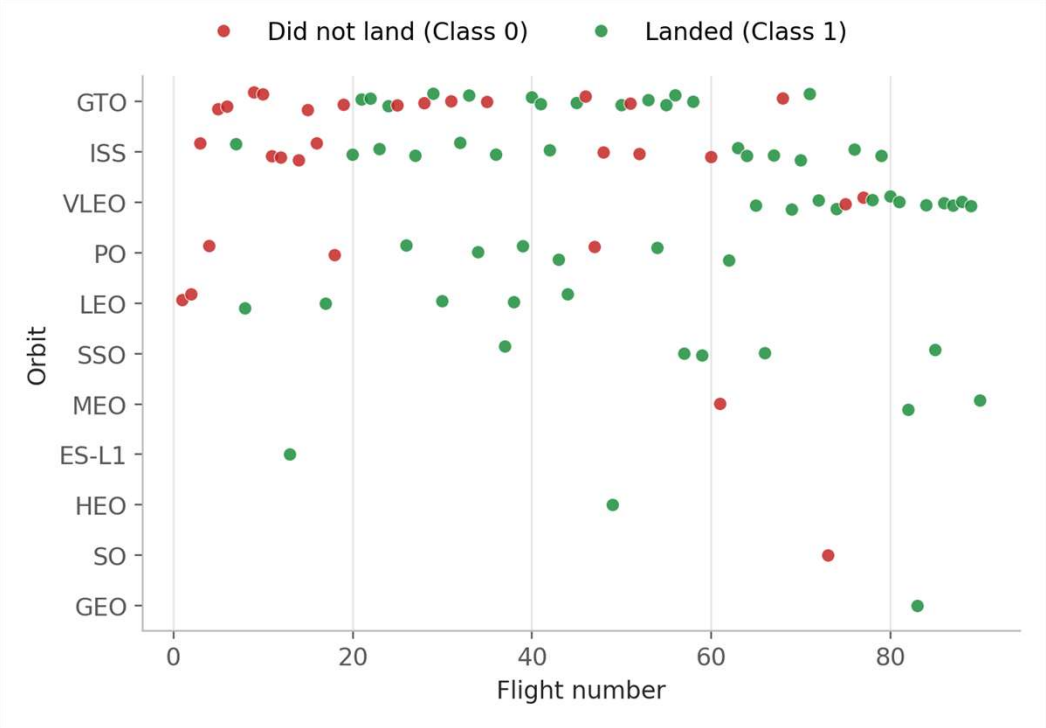
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO have a 100% success rate (but few launches each)
- VLEO (86%) and LEO (71%) are also high
- GTO is the lowest of the common orbits at 52%; SO had 0% (only 1 launch)



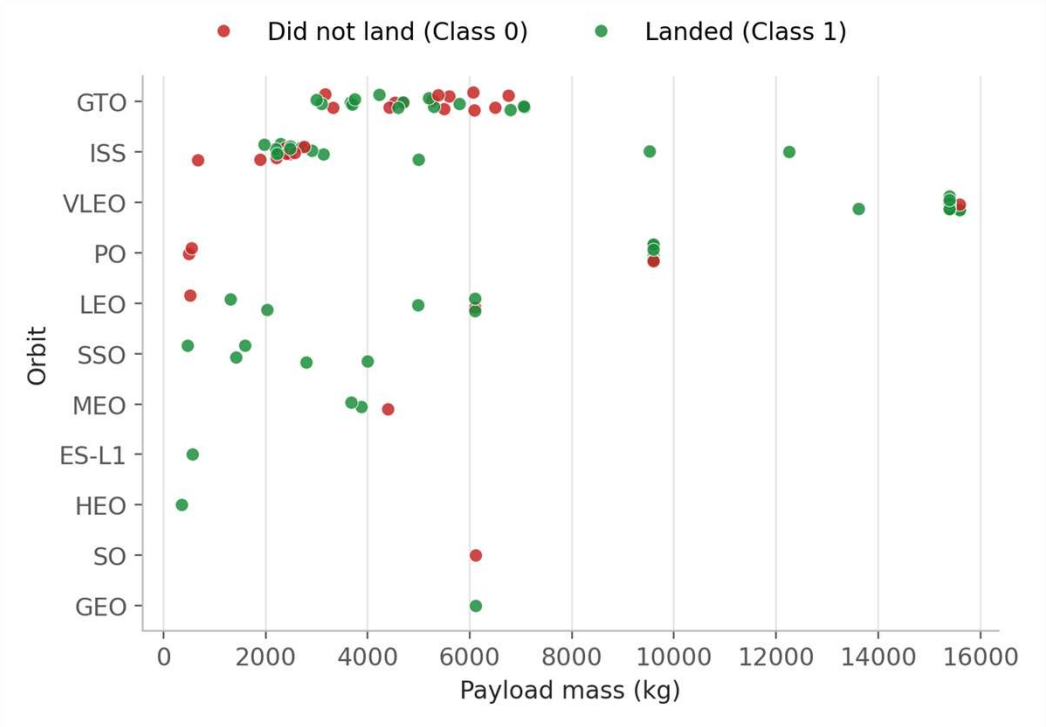
Flight Number vs. Orbit Type

- In LEO, success appears related to the number of flights: later flights succeed more
- In GTO there is no clear relationship between flight number and success
- Newer orbits like VLEO only appear in later flights, which are mostly successful



Payload vs. Orbit Type

- With heavy payloads, successful landings are more common for Polar (PO), LEO and ISS orbits
- For GTO, success and failure are mixed across payloads, so the two are hard to tell apart



Launch Success Yearly Trend

- No successful landings from 2010 to 2013
- The success rate kept increasing from 2013 to 2020 (with a dip in 2018)
- By 2019–2020 about 85–90% of first stages landed successfully



All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

DISTINCT returns each launch site once. SpaceX used four sites: three in Florida and one in California.

Launch Site Names Begin with 'CCA'

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Launch site	Payload	Mass (kg)	Orbit	Customer	Landing outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats…	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	No attempt

These are the earliest Falcon 9 v1.0 flights, all from Cape Canaveral: none of them landed the first stage.

Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

total_payload (kg)
45,596

Boosters launched for NASA's Commercial Resupply Services carried 45,596 kg in total.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

avg_payload (kg)
2,928.4

Mean payload mass for booster version F9 v1.1: about 2,928 kg per launch.

First Successful Ground Landing Date

```
SELECT MIN(Date) AS first_success
FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

first_success
2015-12-22

The first successful landing on a ground pad happened on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Four F9 FT boosters landed on a drone ship carrying 4,000–6,000 kg.

Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) AS total
FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success (with trailing space)	1
Success (payload status unclear)	1

Mission outcomes are almost always successful: 100 successes and only 1 failure. (One 'Success' value has a trailing space in the data, so it shows as a separate row.)

Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4

Booster_Version
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

All 12 are Falcon 9 Block 5 boosters, the most powerful and most reused version.

2015 Launch Records

```
SELECT substr(Date,6,2) AS month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5) = '2015';
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

In 2015 there were two failed drone ship landings, in January and April, both F9 v1.1 boosters from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS cnt FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY cnt DESC;
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

'No attempt' was most common in this early period. Drone ship landings were split evenly: 5 successes and 5 failures.



Section 3

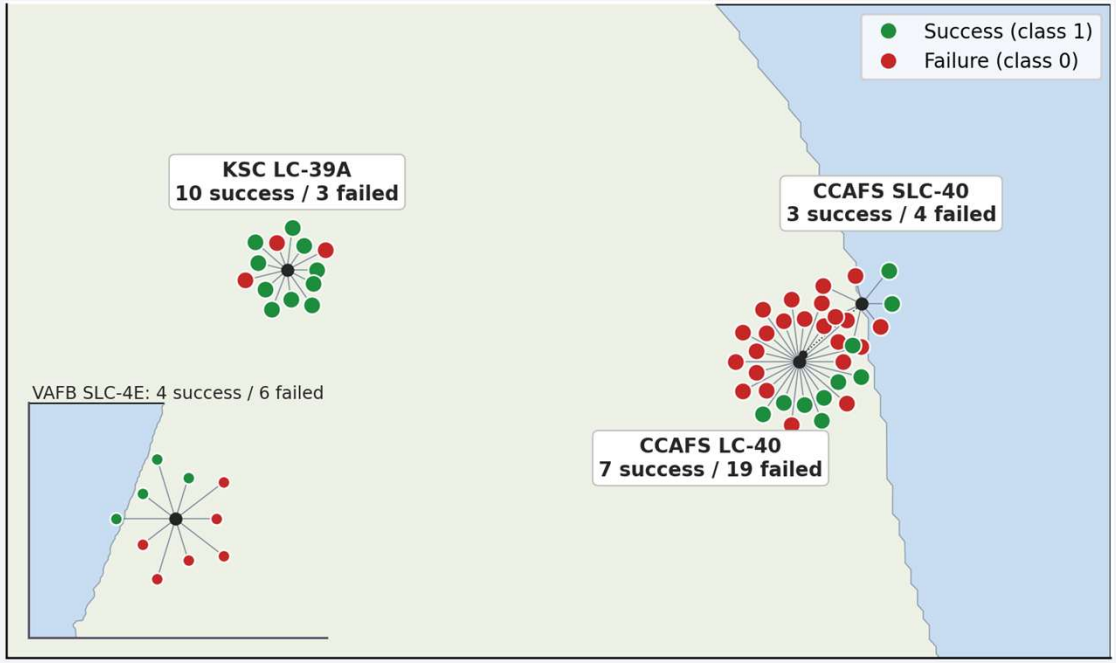
Launch Sites Proximities Analysis

All SpaceX Launch Sites on a Map



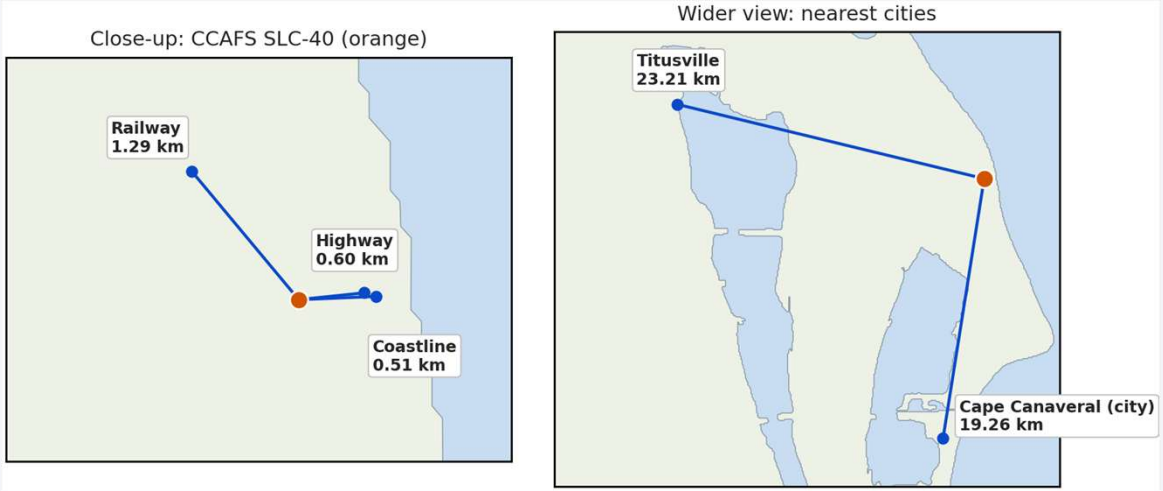
- All four launch sites are in the United States: three in Florida, one in California
- Sites are as close to the equator as possible within the US; the Earth's rotation gives rockets an extra speed boost there
- Every site sits right on the coast, so rockets fly over the ocean and debris falls away from people

Launch Outcomes by Site (Colour-Coded)

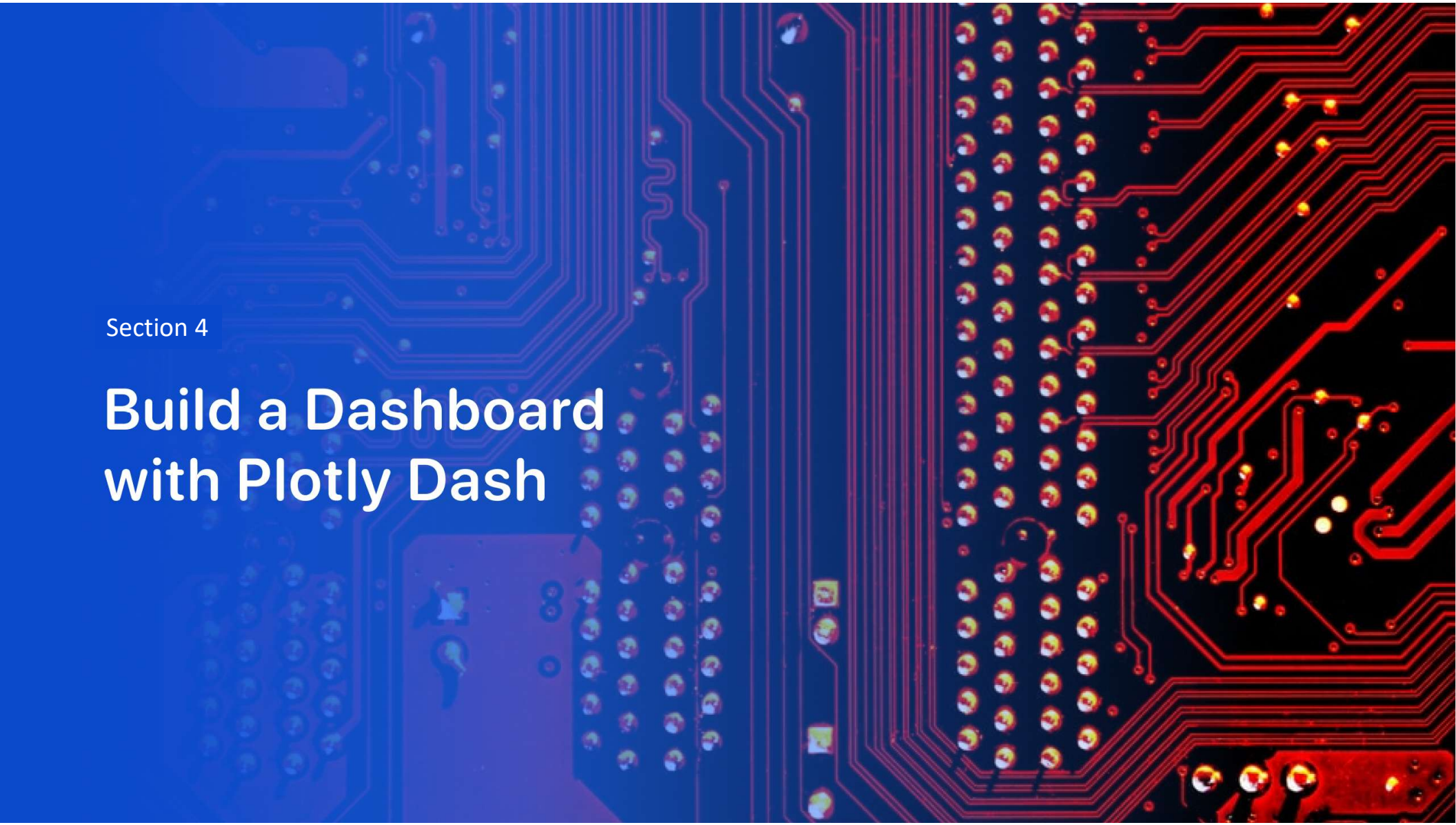


- Green markers = successful landings, red = failures
- Marker clusters show the number of launches at each site
- KSC LC-39A has a high share of green markers, so it has a high success rate
- CCAFS LC-40 has many launches but more red markers from early flights

Launch Site Proximity: Coastline, Railway, Highway, City



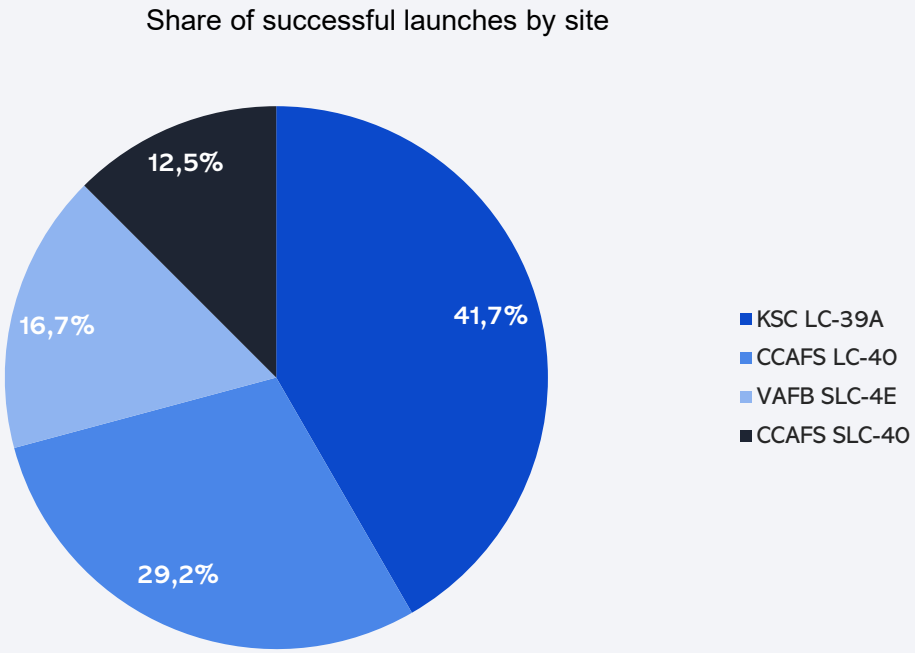
- CCAFS SLC-40 is very close to the coastline (0.51 km), a highway (0.60 km) and a railway (1.29 km)
- Railways and highways are needed to move heavy rocket parts and fuel to the pad
- The nearest cities are much farther away: Cape Canaveral 19.26 km, Titusville 23.21 km
- Sites keep a safe distance from cities in case of an accident, and launch over the ocean



Section 4

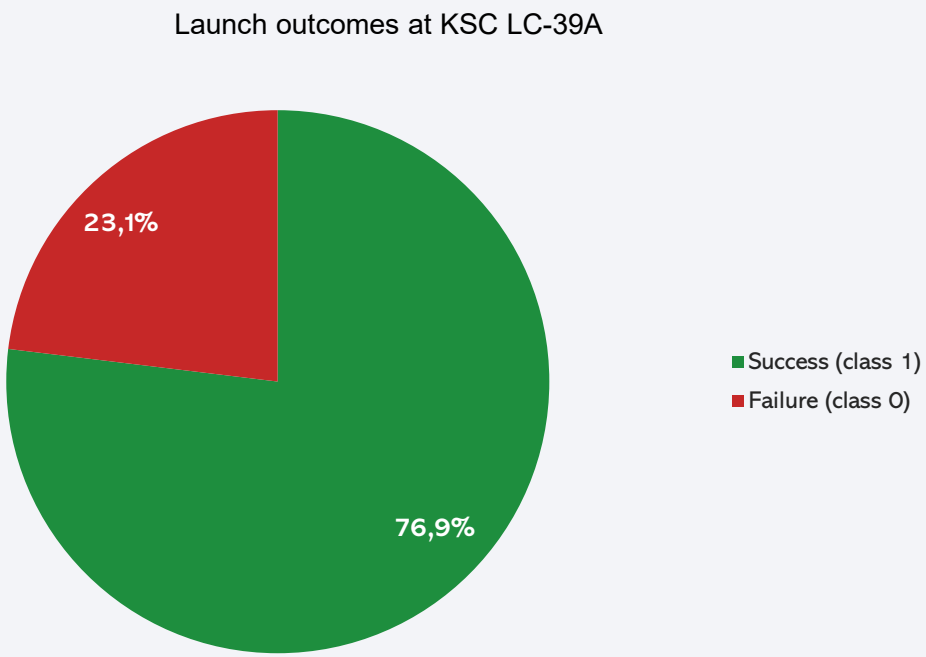
Build a Dashboard with Plotly Dash

Launch Success Count for All Sites



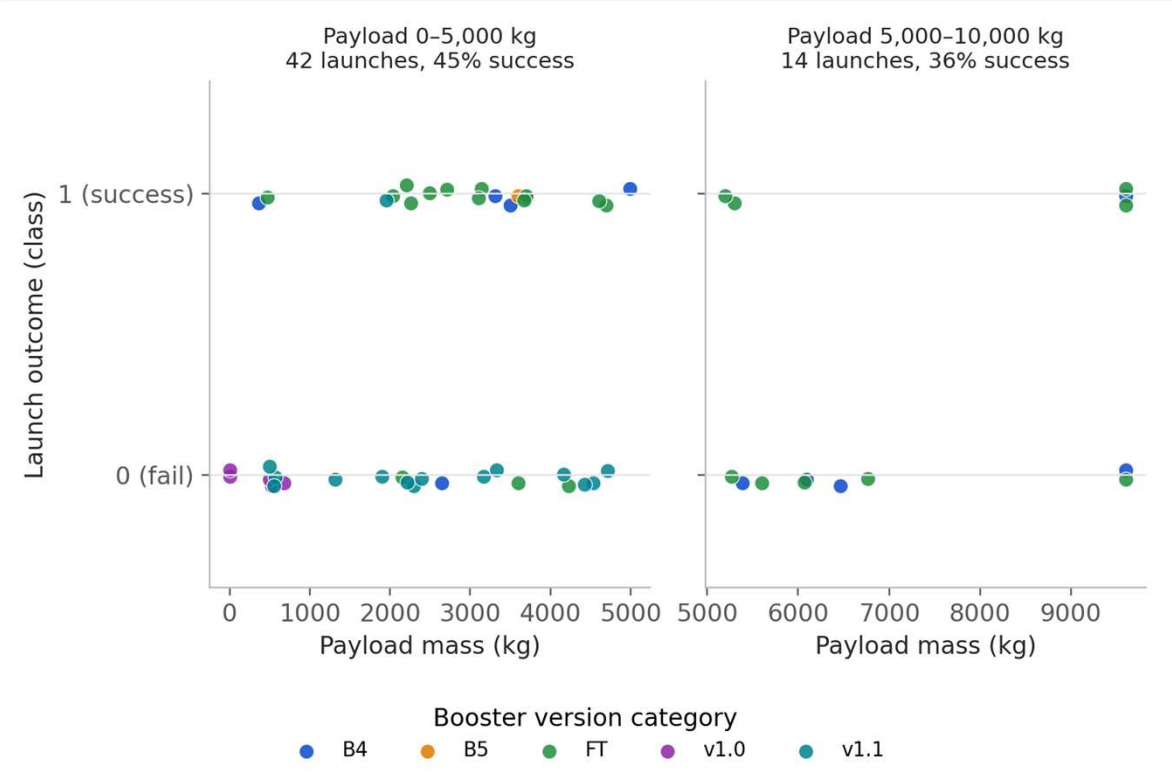
- KSC LC-39A has the most successful launches: 41.7% of all successes
- CCAFS LC-40 follows with 29.2%, VAFB SLC-4E with 16.7%
- CCAFS SLC-40 has the fewest successes at 12.5%

KSC LC-39A: Highest Launch Success Ratio



- KSC LC-39A has the highest success ratio of all sites
- 76.9% of its launches succeeded (10 of 13), and 23.1% failed

Payload vs. Launch Outcome for All Sites



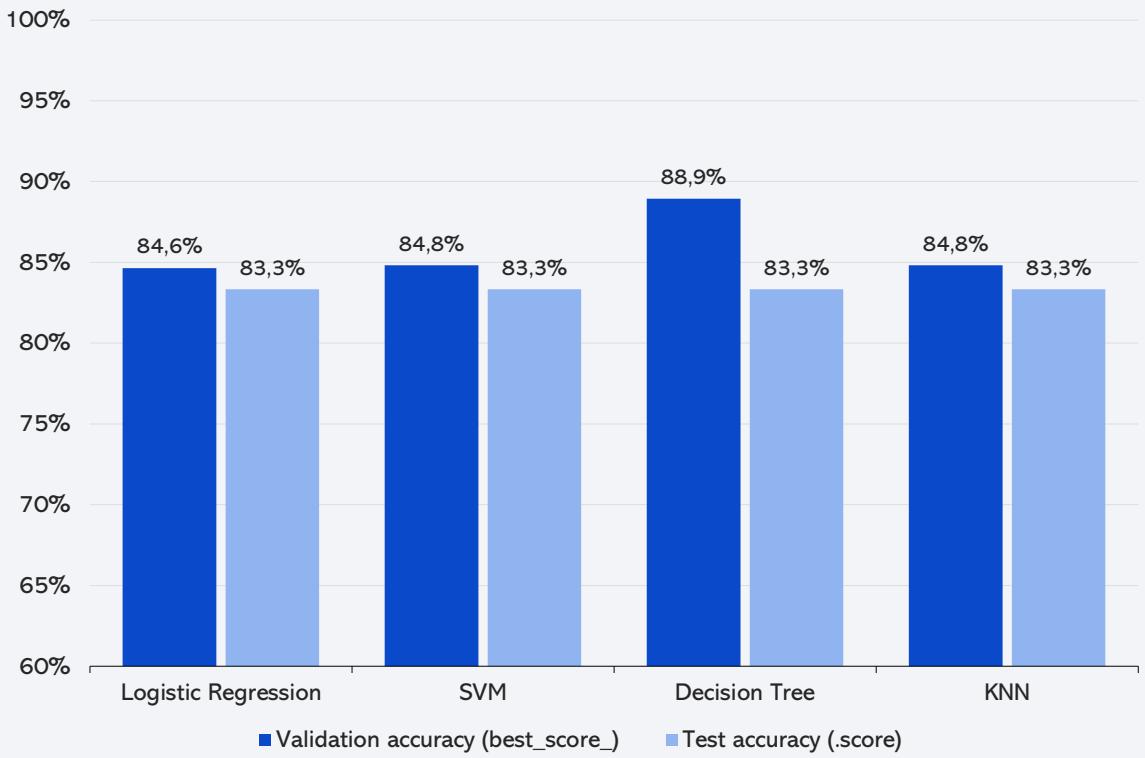
- Payloads of 2,000–5,500 kg have the highest success rate (55%)
- Light (under 2,000 kg: 23%) and heavy (over 5,500 kg: 30%) payloads succeed less often
- Booster version FT has the most successes (16 of 24); v1.0 (0 of 5) and v1.1 (1 of 15) mostly failed

Section 5

Predictive Analysis (Classification)

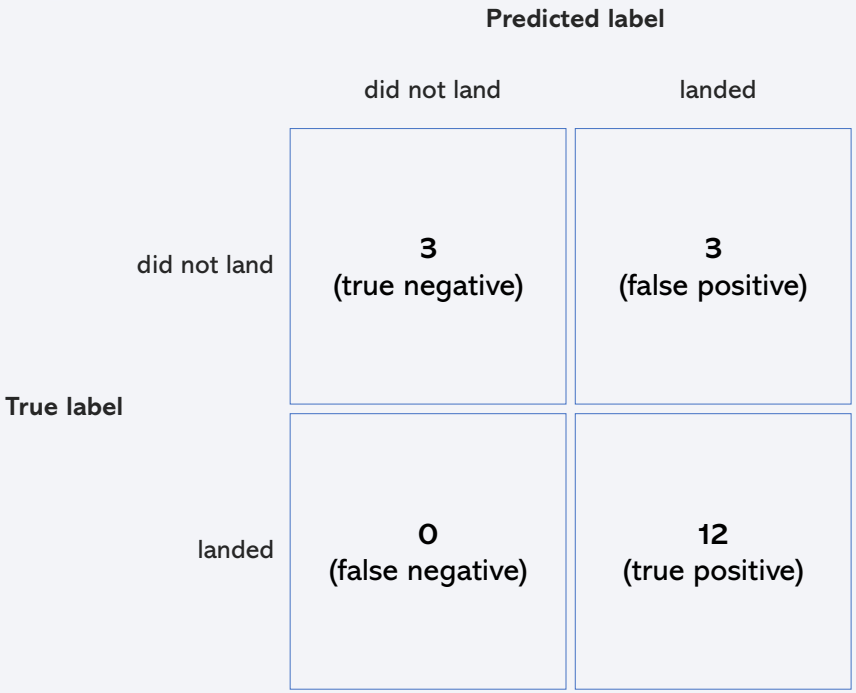


Classification Accuracy



- All four models reach 83.3% accuracy on the 18 test samples
- The Decision Tree has the highest validation accuracy (about 88.9%), so it is the best model
- The test set is small, so the models can't be fully separated on test data alone

Confusion Matrix

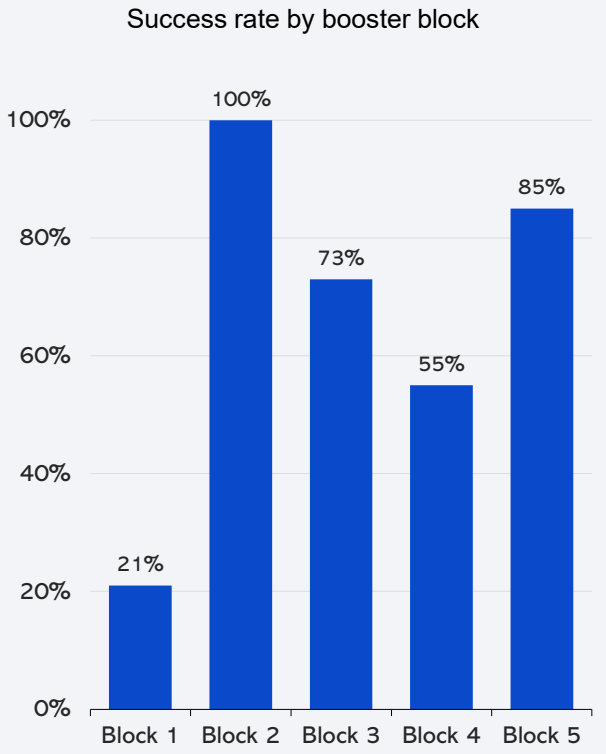


Decision Tree on the 18 test samples (the other models produce the same matrix):

- 12 true positives: landings correctly predicted
- 3 true negatives: failures correctly predicted
- 3 false positives: the main problem. The model predicted a landing that did not happen
- 0 false negatives

Innovative Insight: Booster Hardware and Reuse

- Launches flown without grid fins and legs (18) never landed: they were planned as expendable, so the outcome can be known before launch
- Boosters that had already flown at least once landed about 87% of the time (52 of 60), compared with 27% (8 of 30) for first flights
- Each newer booster generation (Block) landed more reliably, ending at 85% for Block 5 across 39 launches
- For a competitor: if a launch uses a reused Block 5 booster with legs and grid fins, expect a landing, and price against SpaceX's ~$62M launch cost



Conclusions

Experience matters

The more flights a site has completed, the higher its success rate. Success rose from 0% (2010–2013) to around 85–90% by 2019–2020.

Orbit matters

ES-L1, GEO, HEO and SSO had 100% success; GTO was the hardest common orbit at about 52%.

KSC LC-39A leads

It has the most successful launches (41.7%) and the highest success ratio (76.9%).

Location

All sites are near the equator (as far as the US allows) and the coast, close to transport links but away from cities.

Prediction works

All tuned models reach 83.3% test accuracy; the Decision Tree is best on validation accuracy. False positives are the main weakness.

Appendix

Key code snippets

```
# Landing label (data wrangling)
landing_class = [0 if o in bad_outcomes else 1 for o in df['Outcome']]
df['Class'] = landing_class

# One-hot encoding (feature engineering)
features_one_hot = pd.get_dummies(features, columns=['Orbit','LaunchSite','LandingPad','Serial'])
features_one_hot = features_one_hot.astype('float64')

# Model tuning (example: decision tree)
X = preprocessing.StandardScaler().fit_transform(X)
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)
tree_cv = GridSearchCV(tree, parameters, cv=10).fit(X_train, Y_train)
print(tree_cv.best_params_, tree_cv.best_score_, tree_cv.score(X_test, Y_test))
```

GitHub repository: https://github.com/dorotheemorandgrondin/applied-datascience-training

Thank you!

